

1(a)	<table><tr><th>compound / structure</th><th>location</th></tr><tr><td>acetylcholine</td><td>C and D ;</td></tr><tr><td>voltage-gated channel</td><td>A ;</td></tr><tr><td>receptor protein</td><td>B ;</td></tr><tr><td>acetylcholinesterase</td><td>C ;</td></tr></table>	compound / structure	location	acetylcholine	C and D ;	voltage-gated channel	A ;	receptor protein	B ;	acetylcholinesterase	C ;	4
compound / structure	location											
acetylcholine	C and D ;											
voltage-gated channel	A ;											
receptor protein	B ;											
acetylcholinesterase	C ;											
1(b)	any two from: 1 ref. to ion (transport) ; 2 open / closes, when, voltage / depolarisation / charge / electrical potential, changes ; A arrival of an action potential 3 detail ; e.g. ion specificity / hydrophilic pore / transmembrane / (made of) protein	2										
1(c)	any three from: 1 breaks down acetylcholine ; 2 (so) acetylcholine leaves, binding site / receptor / B ; ora 3 depolarisation stops in <u>post-synaptic membrane</u> / B ; 4 stops continuous action potentials (in post-synaptic membrane) ; 5 recycle ACh / described ;	3										

Question	Answer	Marks
1(d)	any two from: 1 one-way transmission ; 2 interconnection of nerve pathways ; 3 AVP ; e.g. memory / learning integration of impulses	2

Question	Answer	Marks
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